# Chapter 3. DERIVATIVES

(Page 213-319, Calculus Volume 1)



MAE101

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# DERIVATIVES

## 3.1 Defining the Derivatives

In this section, we will learn:
How the derivative can be interpreted as a rate of change in any of the sciences or engineering.

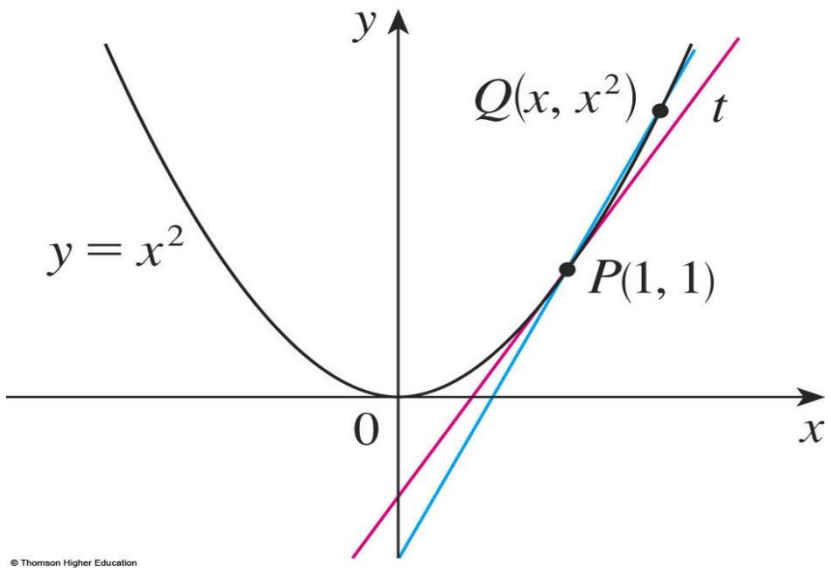
# THE TANGENT PROBLEM

## Example 1

Find an equation of the tangent line to the parabola $y = x^2$ at the point $P(1,1)$.

We will be able to find an equation of the tangent line as soon as we know its slope $m$.

$$m_{PQ} = \frac{x^2 - 1}{x - 1}$$



# THE TANGENT PROBLEM

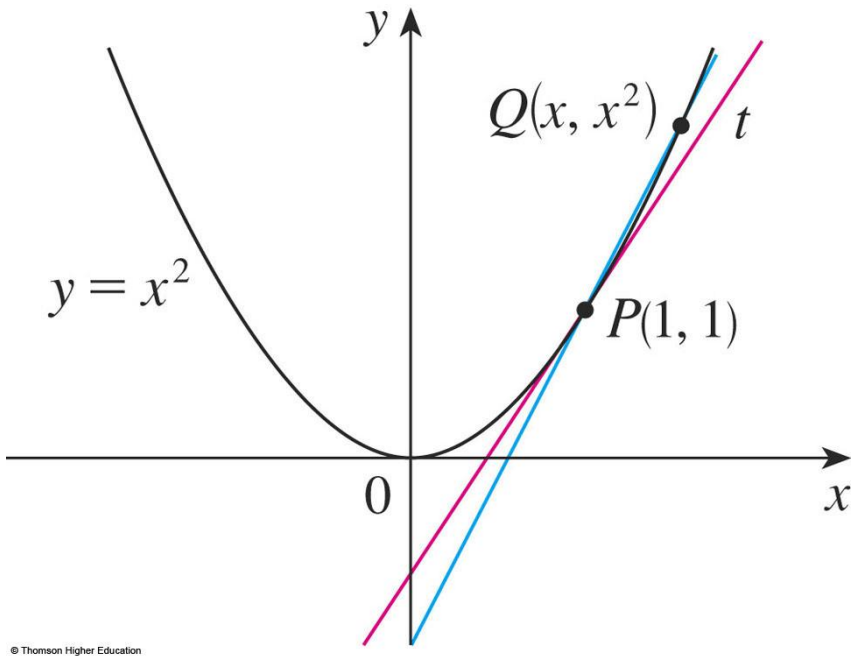
## Example 1

The slope of the tangent line is said to be the limit of the slopes of the secant lines.

$$\lim_{Q \to P} m_{PQ} = m \qquad \lim_{x \to 1} \frac{x^2 - 1}{x - 1} = 2$$

The equation of the tangent line through (1, 1) as:

$$y = 2x - 1$$




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# TANGENTS

## Definition

The tangent line to the curve $y = f(x)$ at the point $P(a, f(a))$ is the line through $P$ **with slope**

$$m = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}$$

provided that this limit exists.

# THE VELOCITY PROBLEM

## Example 2

Investigate the example of a falling ball.

- Suppose that a ball is dropped from the upper observation deck of the CN Tower in Toronto, 450 m above the ground.
- Find the velocity of the ball after 5 seconds.



# THE VELOCITY PROBLEM

## Example 2

If the distance fallen after $t$ seconds is denoted by $s(t)$ and measured in meters, then Galileo's law is expressed by the following equation.

$$s(t) = 4.9t^2$$

# THE VELOCITY PROBLEM

$$\text{average velocity} = \frac{\text{change in position}}{\text{time elapsed}}$$

$$= \frac{s(5.1) - s(5)}{0.1} = 49.49 \text{ m/s}$$

Thus, the (instantaneous) velocity after 5 s is:

$$v = 49 \text{ m/s}$$

| Time interval | Average velocity (m/s) |
|---|---|
| $5 \leqslant t \leqslant 6$ | 53.9 |
| $5 \leqslant t \leqslant 5.1$ | 49.49 |
| $5 \leqslant t \leqslant 5.05$ | 49.245 |
| $5 \leqslant t \leqslant 5.01$ | 49.049 |
| $5 \leqslant t \leqslant 5.001$ | 49.0049 |

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# VELOCITIES

## Definition

We define the **velocity** (or instantaneous velocity) $v(a)$ **at time $t = a$** to be the limit of these average velocities:

$$v(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

# DERIVATIVES

## Definition

The derivative of a function $f$ at a number $a$, denoted by $f'(a)$, is:

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

if this limit exists. Or

$$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}$$

DERIVATIVES

# 3.2 The Derivative as a Function

# DERIVATIVES

In the preceding section, we considered the derivative of a function *f* at a fixed number *a*:

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

If we replace *a* in Equation 1 by a variable *x*, we obtain:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

# OTHER NOTATIONS

Some common alternative notations for the derivative are as follows:

$$f'(x) = y' = \frac{dy}{dx} = \frac{df}{dx} = \frac{d}{dx} f(x) = Df(x) = D_x f(x)$$

The symbols $D$ and $d/dx$ are called differentiation operators.

The symbol $dy/dx$ is called Leibniz notation

# OTHER NOTATIONS

If we want to indicate the value of a derivative $dy/dx$ in Leibniz notation at a specific number a, we use the notation

$$\left.\frac{dy}{dx}\right|_{x=a} \quad \text{or} \quad \left.\frac{dy}{dx}\right]_{x=a}$$

which is a synonym for $f'(a)$.

# OTHER NOTATIONS

## Definition

A function $f$ is differentiable at $a$ if $f'(a)$ exists.

It is differentiable on an open interval D if it is differentiable at every number in the interval D.

# HOW CAN A FUNCTION FAIL TO BE DIFFERENTIABLE?

## Theorem

*If f is differentiable at a, then f is continuous at a.*

$\Rightarrow$ This theorem states that, if $f$ is not continuous at $a$, then $f$ is not differentiable at $a$.

# HIGHER DERIVATIVES

If $f$ is a differentiable function, then its derivative $f'$ is also a function.

So, $f'$ may have a derivative of its own, denoted by $(f')' = f''$.

This new function $f''$ is called the second derivative of $f$.

$$\frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d^2y}{dx^2}$$

# HIGHER DERIVATIVES

The process can be continued.

- In general, the *n*th derivative of *f* is denoted by $f^{(n)}$ and is obtained from *f* by differentiating *n* times.
- If $y = f(x)$, we write:

$$y^{(n)} = f^{(n)}(x) = \frac{d^n y}{dx^n}$$

# DERIVATIVES

## 3.3 Differentiation Rules

## 3.4 Derivatives of Trigonometric functions

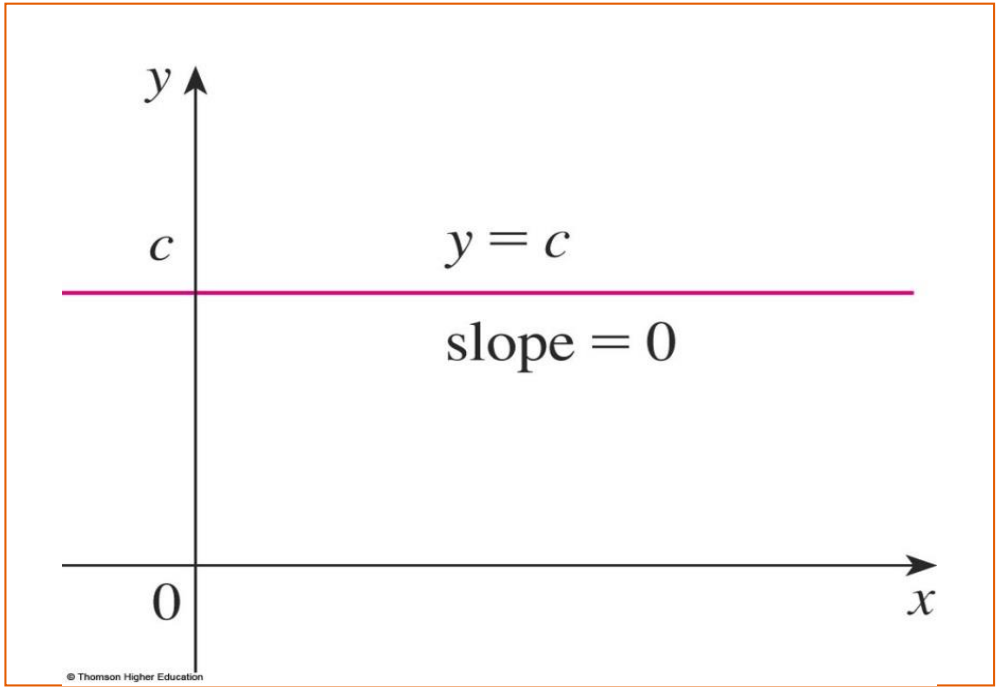
## 3.5 Derivatives of Exponential and Logarithmic functions

# 3.3 DIFFERENTIATION RULES

## Constant function

In Leibniz notation, we write this rule as follows.

$$\frac{d}{dx}(c) = 0$$



# 3.3 DIFFERENTIATION RULES

## Differentiation formulas

Here's a summary of the differentiation formulas we have learned so far.

$$\frac{d}{dx}(c) = 0 \qquad \frac{d}{dx}\left(x^n\right) = nx^{n-1}$$

$$(cf)' = cf' \qquad (f+g)' = f'+g' \qquad (f-g)' = f'-g'$$

$$(fg)' = fg'+gf' \qquad \left(\frac{f}{g}\right)' = \frac{gf'-fg'}{g^2}$$

# 3.5 DERIVATIVES OF TRIGONOMETRIC FUNCTIONS

$$\frac{d}{dx}(\sin x) = \cos x$$

$$\frac{d}{dx}(\cos x) = -\sin x$$

$$\frac{d}{dx}(\tan x) = \sec^2 x$$

$$\frac{d}{dx}(\cot x) = -\csc^2 x$$

$$\frac{d}{dx}(\sec x) = \sec x \tan x$$

$$\frac{d}{dx}(\csc x) = -\csc x \cot x.$$

# 3.9 DERIVATIVES OF EXPONENTIAL AND LOGARITHMIC FUNCTIONS

1. $\frac{d}{dx}(e^x) = e^x$
2. $\frac{d}{dx}(\ln x) = \frac{1}{x}$
3. $\frac{d}{dx}(b^x) = b^x \ln b$
4. $\frac{d}{dx}(\log_b x) = \frac{1}{x \ln b}$

# TANGENT AND NORMAL LINES

## Example 12

Find equations of the **tangent line** and **normal line** to the curve

$$y = \sqrt{x}/(1+x^2)$$

at the point (1, ½).

REMARK:

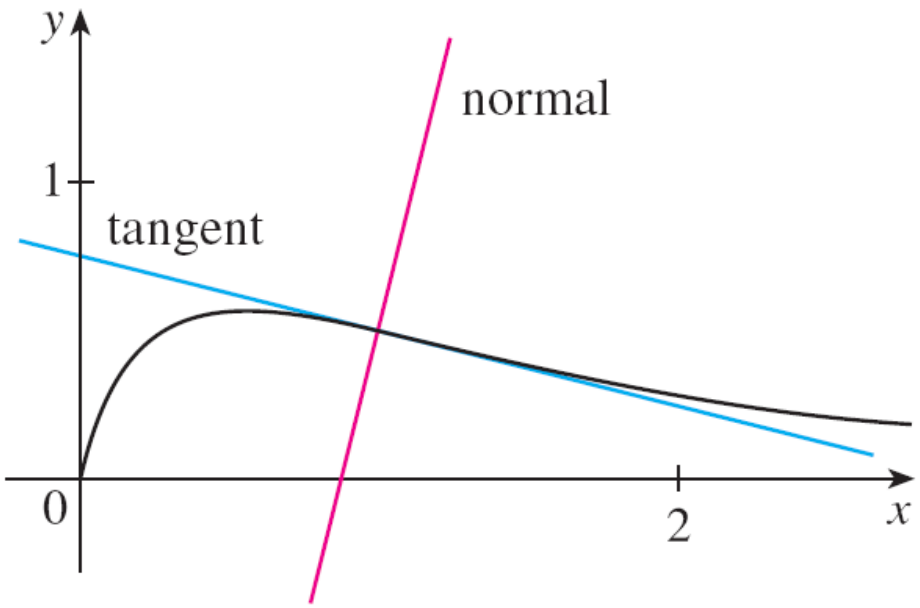
Tangent line $(\Delta)$ equation at $M(x;y) \in (C)$: $y = f(x)$ defined as

$$(\Delta): y = f'(x_0)(x - x_0) + f(x_0)$$

# TANGENT AND NORMAL LINES

Tangent line of the graph of this curve at (1,1/2) is:

$$y = -\frac{1}{4}x + \frac{3}{4}$$




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# DERIVATIVES

## 3.4 Derivatives as Rates of change

# DERIVATIVES

## Definition

Let $s(t)$ be a function giving the position of an object at time $t$.

The velocity of the object at time $t$ is given by $v(t) = s'(t)$.

The speed of the object at time $t$ is given by $|v(t)|$.

The acceleration of the object at $t$ is given by $a(t) = v'(t) = s''(t)$.

# RATES OF CHANGE

## Example

Let $D(t)$ be the US national debt at time $t$. The table gives approximate values of this function by providing end-of-year estimates, in billions of dollars, from 1980 to 2000. Interpret and estimate the value of $D'(1990)$.

| $t$ | $D(t)$ |
|---|---|
| 1980 | 930.2 |
| 1985 | 1945.9 |
| 1990 | 3233.3 |
| 1995 | 4974.0 |
| 2000 | 5674.2 |

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The derivative $D'(1990)$ means the rate of change of $D$ with respect to $t$ when t =1990, that is, the rate of increase of the national debt in 1990.

# RATES OF CHANGE

## Example

By Equation 5, $$D'(1990) = \lim_{t \to 1990} \frac{D(t) - D(1990)}{t - 1990}$$

So, we compute and tabulate values of the difference quotient as follows.

| $t$ | $\frac{D(t) - D(1990)}{t - 1990}$ |
|---|---|
| 1980 | 230.31 |
| 1985 | 257.48 |
| 1995 | 348.14 |
| 2000 | 244.09 |

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| $t$ | $D(t)$ |
|---|---|
| 1980 | 930.2 |
| 1985 | 1945.9 |
| 1990 | 3233.3 |
| 1995 | 4974.0 |
| 2000 | 5674.2 |

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# RATES OF CHANGE

## Example

We estimate that the rate of increase of the national debt in 1990 was the **average of these two numbers**—namely $D'(1990) \approx 303$ billion dollars per year.

| $t$ | $\dfrac{D(t) - D(1990)}{t - 1990}$ |
|---|---|
| 1980 | 230.31 |
| 1985 | 257.48 |
| 1995 | 348.14 |
| 2000 | 244.09 |

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DERIVATIVES

# 3.6

# The chain rule

# THE CHAIN RULE

If $g$ is differentiable at $x$ and $f$ is differentiable at $g(x)$, the composite function $F = f \circ g$ is differentiable at $x$ and $F'$ is given by the product:

$$F'(x) = f'(g(x)) \cdot g'(x)$$

– In Leibniz notation, if $y = f(u)$ and $u = g(x)$ are both differentiable functions, then:

$$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}$$

| Let $f(x)=g(\sin 3x)$. Find $f'$ in terms of g'. | |
|---|---|
| A | $3\cos 3x\, g'(x)$ |
| B | $3\cos 3x\, g'(\sin 3x)$ |
| C | $\cos 3x\, g'(\sin 3x)$ |

| Suppose $h(x)=f(g(x))$ and $f(2)=3$, $g(2)=1$, $g'(2)=1$, $f'(2)=2$, $f'(1)=5$. Find $h'(2)$. | |
|---|---|
| A | 1 |
| B | 2 |
| C | 3 |
| D | 4 |
| E | 5 |

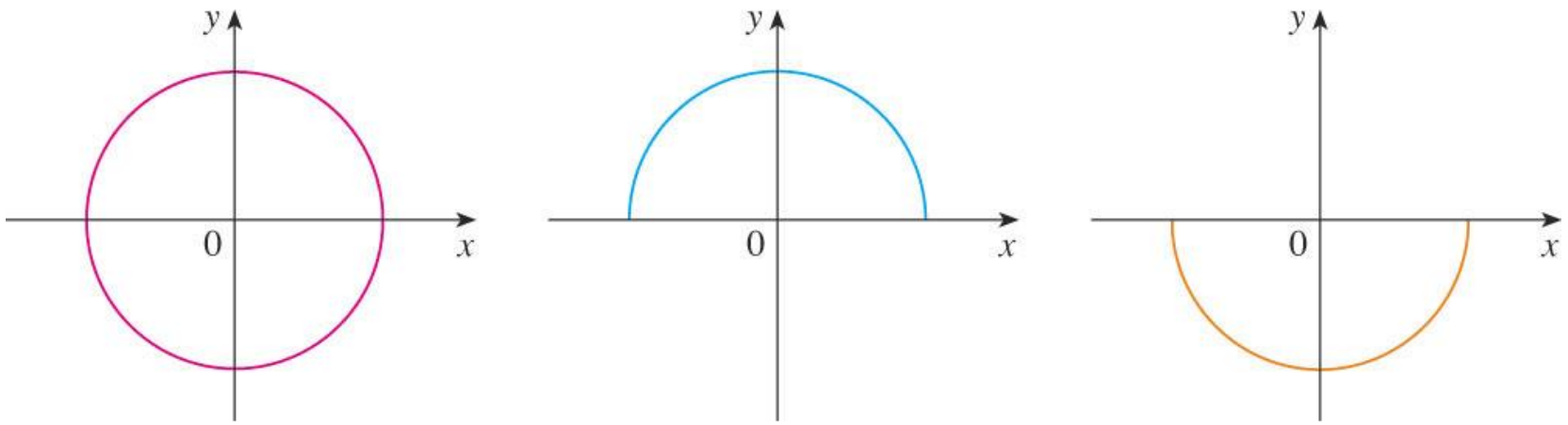
DERIVATIVES

# 3.6

# Implicit Differentiation

# IMPLICIT DIFFERENTIATION

The graphs of $f$ and $g$ are the upper and lower semicircles of the circle $x^2 + y^2 = 25$.




(a) $x^2 + y^2 = 25$ (b) $f(x) = \sqrt{25 - x^2}$ (c) $g(x) = -\sqrt{25 - x^2}$

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# IMPLICIT DIFFERENTIATION METHOD

Instead, we can use the method of implicit differentiation.

- This consists of differentiating both sides of the equation with respect to $x$ and then solving the resulting equation for $y'$.

# IMPLICIT DIFFERENTIATION

## Example 1

a. If $x^2 + y^2 = 25$, find $\dfrac{dy}{dx}$

b. Find an equation of the tangent to the circle $x^2 + y^2 = 25$ at the point $(3, 4)$.

$$\frac{d}{dx}(x^2 + y^2) = \frac{d}{dx}(25)$$

$$\frac{d}{dx}(x^2) + \frac{d}{dx}(y^2) = 0$$

# IMPLICIT DIFFERENTIATION

## Example 1 a

Remembering that $y$ is a function of $x$ and using the Chain Rule, we have:

$$\frac{d}{dx}(y^2) = \frac{d}{dy}(y^2)\frac{dy}{dx} = 2y\frac{dy}{dx}$$

$$2x + 2y\frac{dy}{dx} = 0$$

Then, we solve this equation for $\frac{dy}{dx}$: $\frac{dy}{dx} = -\frac{x}{y}$

# IMPLICIT DIFFERENTIATION

## E. g. 1 b—Solution 1

At the point (3, 4) we have $x = 3$ and $y = 4$.

So, $\frac{dy}{dx} = -\frac{3}{4}$

Thus, an equation of the tangent to the circle at (3, 4) is: $y - 4 = -\frac{3}{4}(x - 3)$ or $3x + 4y = 25$.

# IMPLICIT DIFFERENTIATION

**Example 4**

Find $y''$ if $x^4 + y^4 = 16$.

Differentiating the equation implicitly with respect to $x$, we get $4x^3 + 4y^3y' = 0$.

# IMPLICIT DIFFERENTIATION

## E. g. 4—Equation 3

Solving for $y'$ gives:

$$y' = -\frac{x^3}{y^3}$$

# IMPLICIT DIFFERENTIATION

## Example 4

To find $y''$, we differentiate this expression for $y'$ using the Quotient Rule and remembering that $y$ is a function of $x$:

$$y'' = \frac{d}{dx}\left(-\frac{x^3}{y^3}\right) = -\frac{y^3(d/dx)(x^3) - x^3(d/dx)(y^3)}{(y^3)^2}$$

$$= -\frac{y^3 \cdot 3x^2 - x^3(3y^2y')}{y^6}$$

# IMPLICIT DIFFERENTIATION

## Example 4

If we now substitute Equation 3 into this expression, we get:

$$y'' = -\frac{3x^2y^3 - 3x^3y^2\left(-\frac{x^3}{y^3}\right)}{y^6}$$

$$= -\frac{3(x^2y^4 + x^6)}{y^7} = -\frac{3x^2(y^4 + x^4)}{y^7}$$

# IMPLICIT DIFFERENTIATION

## Example 4

However, the values of $x$ and $y$ must satisfy the original equation $x^4 + y^4 = 16$.

So, the answer simplifies to:

$$y'' = -\frac{3x^2(16)}{y^7} = -48\frac{x^2}{y^7}$$